

Streamline your business through Quality Management

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Quality Management

What we want to cover in this session is

What is Quality

Origins of Quality systems

Some regulator requirements

Identifying some roadblocks to a streamlined business

How can we stretch what we have to make it work.

Quality Management

It's time to get you involved in your own business

What is Quality

There are a number of definitions, but they all reflect the same message.

This is one of the many:-

Quality is the degree to which an object or entity (e.g., process, product, or service) satisfies a specified set of attributes or requirements.

Customer Satisfaction

Conformance to requirements

Fitness for purpose

Fitness for Purpose

What restaurant has the highest quality?,

A world-renowned (3 star) Michelin star certified restaurant

Or

Mc Donalds, a fast-food chain

Given only that information, the Michelin Star of course

Scenerio

We want to celebrate our 4-year-old son's birthday and we want him to have a "quality" day, so out of those two, where do we go?

McDonalds of course (fitness for purpose)

Now don't get me wrong, McDonalds restaurant have an excellent quality assurance and management system in place.

Remember that definition we mention previously,

Quality is the degree to which an object or entity (e.g., process, product, or service) satisfies a specified set of attributes or requirements.

Origins of Quality systems

"Quality is built in, not inspected in."

To fully understand this principle requires a historical perspective, particularly a look back to around 1942 during World War 2.

War products were being manufactured by factories that normally produced domestic goods.

All the various manufacturers used different tolerances due to their machinery capabilities.

The rework of each component caused huge delays in assembly.

But a bigger problem arose, in the manufacture of ammunition.

In 1940's Quality was "inspected in" one round out of every 1000 was test fired, if it worked then the rest (999) were shipped to the soldiers.

However, if it failed the whole 999 were sent back to the factory for rework or scrapping.

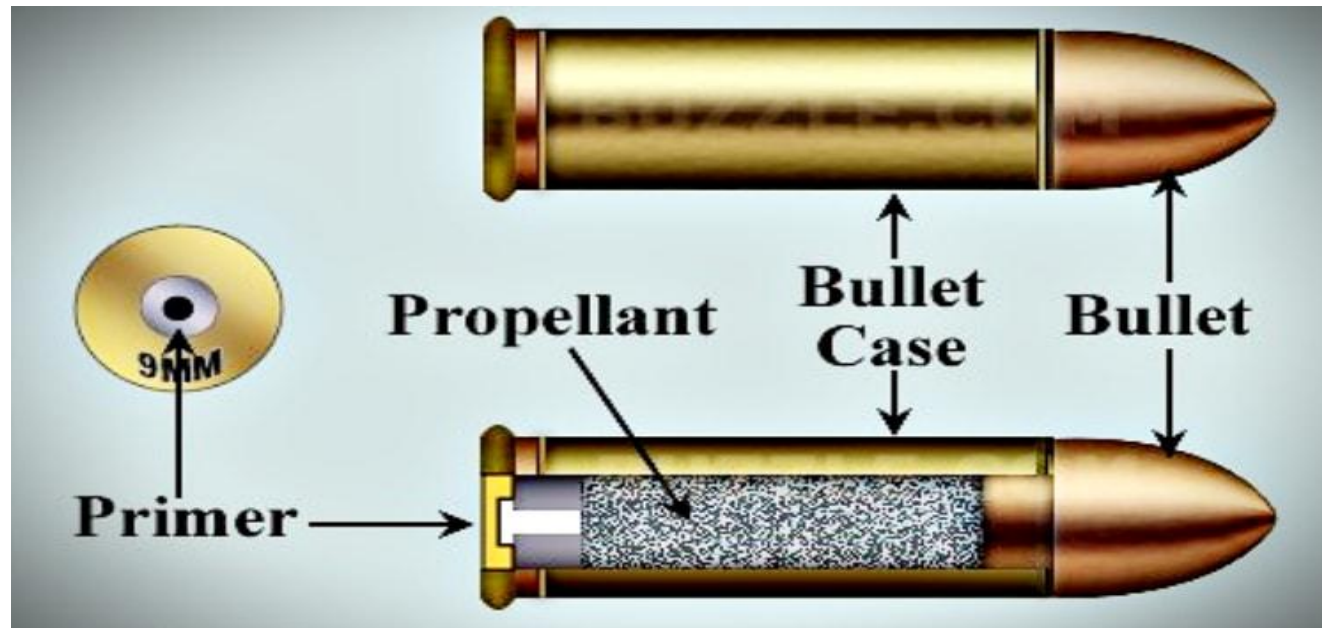
In either case, the test bullet might have been the only one in that 1000 that exhibited that characteristics.

Numerous reports were coming back that there was a high failure rate in the field.

Obviously, there was a need for improvement.

How did they improve?

Bullets are comprised of several components—including cases, primers, powder, and projectiles—often sourced from different suppliers.



Upon arrival at the assembly plant, Each component was subjected to rigorous checks:

cases were weighed and measured according to documented standards,

primers were checked for proper fit,

powder was verified for correct composition,

projectiles were inspected for dimension and weight.

Non-compliant parts were returned to suppliers to maintain quality at every stage.

We have now created a basic form of incoming inspection

A new system emerged: divide the assembly process into sections, assigning each segment responsibility for its output.

The assembly sequence involved verification at each step—installing primers, adding powder, and seating the projectile—with checks performed before advancing to the next stage. Any discrepancies led to immediate rectification before continuation.

No requirement to test 1 in 1000 any more because quality has been built in.

Regulator Quality system requirements

A part 145 workshop has a requirement to have a quality system.

When you look at the regulatory system as a whole you will see many similarities with an ISO9000 type quality management system.

While the quality system required by the regulator are designed to enhance safety, they do very little to assist you or your staff in the everyday business.

Regulator requirements for quality – your exposition document plus a couple more items, all focused on the regulations.

I'm not going to spend time on the regulatory quality system requirements

What is important to say is that many organisations run their whole business to satisfy the regulator.

This is not a healthy way to achieve growth or profit

Document and data control

The regulatory requirement is only about all documents and data relating to the maintenance of Aircraft.

Some examples are

job history

Maintenance Manuals

Component traceability

All necessary to satisfy the regulator but are of no help to us running a business.

Time is money

Our business revolves around labour charge out and any wasted time is lost income for the business

10 minutes here and there can add up to a lot of lost hours per week.

How can a quality system reduce the lost hours

First, we have to identify the lost time.

Quality consists of customer/supplier relationships.

We have two types of customer and suppliers.

External

Internal

External customers are airlines who we repair equipment for, and our external suppliers are the OEM's who we purchase our parts from.

However, the detail is in your business, the internal customers and suppliers.

Supplier / Customer relationship

- It is easy to relate quality with our external customer, they expect a certain outcome and if we provide services that are fit for purpose, then they pay you for those services. Simple.
- Your quality management manual should have a dispute resolution procedure to achieve an outcome in the shortest possible time.

Internal Supplier/Customer

- What about our internal customers and suppliers
- Generally, the most wasted time occurs when there is a breakdown somewhere in that Supplier / Customer relationship
- As an example, we will use a piece of equipment for repair and return to demonstrate this.

Internal Customer - Supplier

The arrangement of customer supplier changes due to the position in the chain of events, let me explain.

Our External Customer, the aircraft owner/Operator(ABC) airlines provides you with a piece of equipment for repair and return.

OK, lets follow this component through

First step

ABC is now the supplier, and you are the Customer.

You both have responsibilities

Supplier, ABC is responsible to provide correctly signed purchase order with sufficient detail about what they require.

Customer, you are responsible to ensure that all of that is provided.
(Incoming Inspection)

Next

You then create a job using the details required and supply it all to the workshop.

Now, you are the supplier, and the workshop is the customer

Suppliers' responsibility is to provide all the details and equipment for repair to the workshop.

The workshop is responsible for ensuring everything required is there and correct. (incoming Inspection)

The workshop carry out the task required and you satisfy the regulator by including the correct documentation, then it is supplied to the store for returning to ABC.

Now the workshop is the supplier, and it has the responsibility to provide the necessary and correct documentation to the store (Despatch)

The Store is now the customer and is responsible for ensuring all the equipment (Serial N0.s etc) and paperwork are correct. (incoming Inspection)

Finally

The store can pack the unit for return to ABC

Now the store is the supplier, responsible for ensuring that all the necessary bits and pieces are correctly packaged and returned to ABC

This doesn't let ABC off the hook, ABC is still the customer and must ensure that what you supplied was complete and correct.(Incoming Inspection)

- Every time the Supplier – Customer changes position there is an opportunity to lose time.
- Example
- Imagine if you are paying your store person overtime to dispatch a backlog of jobs because the workshop, in its roll as supplier, did not supply the correct data to the store for dispatch of one single job.
- This actually happened in my business

- OK, that was a store /workshop issue, but the same speedbumps occur with accounting, e g invoicing. Someone not supplying the correct data.
- These things can happen anywhere within your business.
- It is vitally important that you identify and address these potential problems.

- Start expanding your quality system to include the processes to reduce the possibility of wasted time.
- Don't forget to add any document and data control required for your business data.
- When you audit your quality system, include the business processes and procedures in it and look for continual improvement

- Everyone's business is different; you all have reached a different level of maturity. So. there is no “one size fits all” solution
- It's up to all members of the staff to identify these speedbumps and the best people to ask are, the people doing the job, they know where the pitfalls are.
- So, engage all the staff, Quality is not a management decree, it is a system owned and operated by all the staff.
- Get your staff to document the problems they have and collectively find a solution.

Finally

- Ensure that any resolution to a problem continues to make the “Supplier” responsible for their own output
- Document the resolution, but don’t write a small novel, a few lines will sometimes be sufficient.
- AND
- Make the procedures available to the staff

Thank you